

NELSON KANG

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Education

University of Waterloo

Expected Apr 2030

Bachelor of Computing and Financial Management (CFM)

Activities: Poker Club Executive, Competitive Intramural Volleyball Captain, Training for Ironman 70.3 Muskoka 2027

Technical Skills

Languages: Python, C#, C, C++, SQL, TypeScript, JavaScript, HTML/CSS

Data & Infrastructure: AWS (S3, Athena, EventBridge, CodePipeline), Apache Iceberg, PyIceberg, DuckDB, Kafka, PostgreSQL, SQL Server, Redis, Docker, Terraform

Frameworks & Tools: FastAPI, Next.js, Celery, Modal, Supabase, Pandas, NumPy, Power BI, Postman, Git, Linux

ML / CV: Pytorch, YOLOv8, Roboflow, OpenCV, FFmpeg

Experience

Backend & Data Engineering Co-op

Jun 2026 – Aug 2026

Aquanow

Toronto, ON

- Migrated the company-wide month-end client reconciliation from a legacy JavaScript job to a containerized, **Kafka-triggered Lakeflow** worker (internal job framework) over **Apache Iceberg**, validating **3,800+** account identities against legacy output; root-caused **two long-standing production discrepancies**.
- Designed and built an **AI-agent reconciliation workflow** that classifies breaks by failure mode and auto-investigates unexplained residuals, producing review-ready reports for the finance team.
- Extracted a shared **reconciliation library**, reused it on a second reporting domain alongside a second co-op engineer, and authored the org-level **RFC and ADR** proposing it as a company-wide pattern.
- Diagnosed and fixed **5 critical production bugs** in a live regulatory reporting pipeline (**Python, AWS S3/Athena, PyIceberg**) submitting daily trading activity to **Turkey's securities depository (MKK)**, including a full-day snapshot-boundary offset and an in-asset fee misattribution; restored reconciliation from **0 to all 107 currencies**.

Software Engineer Intern

Jun 2024 – Aug 2024

Ericsson Canada Inc.

Ottawa, ON

- Built **4 C#/SQL Server ETL pipelines** joining multiple Jira endpoints with internal tables into reporting tables, using **temporal tables** for full change-history tracking.
- Audited and catalogued all team pipelines, documenting faulty custom-field mappings in a reference that became the team's standard **debugging resource**; maintained **Power BI** dashboards for trouble-report visibility.

Projects

ClipFarm — Automated Volleyball Highlight Generator

Mar 2026 – Present

Python, FastAPI, Celery, Redis, Modal, TypeScript, YOLOv8-pose, Roboflow, PostgreSQL, Docker, R2

- Founded and lead a team of 4 engineers** building a **multi-signal highlight pipeline**: a custom-trained ball detector bounds rallies by trajectory, audio cheer detection and rally-shape scoring gate highlights, and **YOLOv8-pose** classifies player actions.
- Containerized with **Docker Compose**: **FastAPI** serves auth-gated endpoints, **Celery + Redis** distributes jobs, **Modal** offloads GPU inference; auth via **Supabase JWKS** with **Row-Level Security**.
- Built the **Next.js + TypeScript** frontend: filterable clip grid, inline multi-label correction UI feeding future training, background clip re-cut via Celery, and lazy-loaded thumbnails.
- Built an **autonomous overnight AI agent loop** that clears non-blocking tickets, opens PRs with multi-pass automated review, and reports each morning for human sign-off.

HUME — Belief-Revision Engine for Biology Research Findings

Jul 2026

1st Place, CORTEX BioSciences Track @ Hack the 6ix (\$1,500) · Python, no external libraries

- Built a **deterministic belief-revision engine (no LLM)** that ingests streaming evidence into a **knowledge graph**, sizing each update in **log-odds** by evidence strength and scoping refuted claims to the condition that breaks them.
- Blocked **prompt injection** structurally: untrusted text can never write state, and all updates flow through a fixed schema with provenance-derived weights. Independent **90-attack red-team: zero** text-driven mutations.

Multi-Factor Portfolio Advisor — 1st Place, Market Beat Competition

Nov 2025

Python, Pandas, NumPy, yfinance

- Simulated **\$1M portfolio** returned **5.7%** vs. a 3.7% benchmark, placing **1st of 18 teams**; built a **multi-factor pipeline** (momentum, volatility, trend) with **sector-cap enforcement** and **Sharpe-ratio optimization** under real-world constraints.